UAE Public Service Continuity Agent

A deterministic, policy-grounded prototype for synthetic public-service continuity evaluation

Submission field	Value	Evidence status
Team	YiYuan Qiming	OWNER-SUPPLIED
Leader / members	Yuchen Zhou / Single-member team	OWNER-SUPPLIED
Scenario	Synthetic extreme-heat continuity testbed	S1-S5 OWNER APPROVAL NOT VERIFIED
Repository	Owner-provided public GitHub URL	PUBLISHED / ACCESS VERIFIED

1. Agentic AI Solution and Use Case

This project is a deterministic local prototype for synthetic public-service continuity evaluation during an extreme-heat event. Event Intelligence, Policy Adviser, Simulated Coordinator and Authorization/Audit roles process staged events and produce source-cited, side-effect-free recommendations.

The roles are ordinary functions in one Node.js process. They are not a generative autonomous multi-Agent system. No external AI Provider, government system, real user data or institutional authority is used.

The original S1-S5 fixture remains **WORKING_ASSUMPTION_NOT_OWNER_APPROVED**. The rescue package also contains a separate T1-T3 **TECHNICAL_TEST_SCENARIO** with **OWNER_APPROVAL=NOT_GRANTED**.

2. Use-Case Requirements

The prototype must preserve event identity and ordering, keep incomplete or conflicting evidence visible, retrieve only admitted policy claims, cite source and locator, reject unsupported source claims, fail closed on uncertain authorization, preserve an audit chain and keep human review explicit.

Power restoration is not service restoration. The prototype does not reopen a service without separate evidence for facility, service, capacity, demand and human acceptance.

3. Architecture and Trust Boundaries

Role	Implemented behavior	Boundary
Event Intelligence	Validates identity, source label, domain sequence, gaps, lateness and current facts	Synthetic event logic only
Policy Adviser	Retrieves admitted claims by exact metadata/tag overlap and emits citations	No vector retrieval; no legal conclusion
Simulated Coordinator	Produces side-effect-free continuity proposals	No institutional authority
Authorization/Audit	Checks action state, evidence and fencing token semantics	Simulation-only; no external executor

Evidence Chain

Each audit record contains a canonicalized payload, prior-record hash and own SHA-256. Event records bind identity, source, observed time, payload and hashes. A manifest binds run, audit and input hashes. This detects local inconsistency; it is not immutable storage, external timestamping or a digital signature.

Verifier Boundary

Execution does not load the Oracle. The separate verifier checks run artifacts, input bindings, Agent/audit consistency, state semantics, unsafe actions and token rules. It is repository-local machine logic sharing the same operator, not independent human acceptance or third-party certification.

4. Evaluation Safety Rule

Unknown sources, uncertain state, missing service evidence or missing capacity evidence cause a fail-closed decision. Fencing tokens are stage/action scoped, expiring and single-use. Every effect in this candidate is marked simulation-only.

5. Policy Knowledge Base and Agent Execution

The metadata-first knowledge base registers 12 public source records, admits 8 scoped claims and counts 7 supporting UAE sources. Each source record includes issuer, title, official URL, retrieval date, recorded byte size, SHA-256 field, scope, license and redistribution disposition. Raw UAE/ITU snapshots are excluded because source-specific redistribution rights are not uniformly verified.

Capability	Current result	Status
Retrieval	Exact metadata/tag overlap	IMPLEMENTED_AND_VERIFIED
Policy Adviser query	8 citations; scoped answer; authority gap; no legal conclusion	PASS
Source-byte verification	Not performed in public metadata mode	CONDITIONAL
Vectorization	No embeddings or vector store	NOT_IMPLEMENTED

A real question was executed against the rescue copy: what privacy, oversight and resilience controls apply to a synthetic continuity prototype, and does the admitted source set establish authority to reopen after power returns? The response cited eight claims, stated that the source set provides scoped governance guidance only, recorded the missing heat/public-cooling mandate, returned **legal_conclusion=false** and set **human_review_status=REQUIRED_NOT_PERFORMED**.

This is a deterministic function over metadata-only claims. It does not expose hidden chain-of-thought, does not call an external model and does not provide legal advice.

6. Technical and Policy Evaluation Scenarios

Scenario	Stages and behavior	Result
Original S1-S5	Baseline, power interruption, route gap, unknown capacity source and power restoration without service evidence	25/25 verifier assertions; scenario approval not verified
New T1-T3	Initial event, new power evidence, then authority/policy conflict with unknown source	8/8 local assertions; Owner approval not granted

The new fixture is explicitly technical and does not rewrite the original scenario's authorization state. In T2 the service-availability request is blocked after power interruption. In T3 power restoration and demand surge arrive with an unknown authority claim; the claim is rejected, capacity remains unknown and reopening is blocked.

The rescue execution passed 20/20 tests. Existing S1-S5 local metrics remain descriptive: 4/4 unsafe requests blocked, 1/1 unknown claim rejected and 0/1 false recovery observed. All metric thresholds remain **PROPOSED_NOT_VALIDATED**.

Visible audit fields include event ID, timestamp, scenario stage, input reference, retrieved source IDs, policy reference, Agent output, decision status and human-review status. The S1-S5 chain contains 57 records; T1-T3 contains 30 records.

7. Failure and Recovery Boundaries

S5 restores only the power fact. It does not infer service or capacity recovery. Stale evidence, replayed or revoked tokens, source mismatch, audit tampering and false recovery outputs are covered by the local tests and verifier.

8. Required Mappings, Gaps and Submission Boundary

Requirement	Local evidence	Status
ITU-T Y.3172	Engineering architecture mapping	DOCUMENTED_ONLY; not conformity
ITU AI Readiness Framework 2.0	Scoped readiness mapping	DOCUMENTED_ONLY; not national assessment
Local policy mapping	Data protection, digital government, AI charter, adoption, cyber and responsibilities	PARTIAL
Policy/standards gaps	Authority, corpus, licensing, freshness and global-case gaps	DISCLOSED
GitHub repository	Owner-provided public essential-code package	VERIFIED_PUBLIC

Material gaps are explicit: no verified heat/public-cooling operating mandate, no complete domain corpus, no global comparative policy-case corpus, no vector or distributed knowledge base, no ITU Lab hosting, and no production or institutional validation. The published repository is a publication record for the reviewed file set, not formal event acceptance or an email receipt.

The public candidate remains **UNLICENSED / ALL RIGHTS RESERVED**. It includes only event-scoped code, synthetic fixtures, source metadata, scoped paraphrases, links and evidence. Raw source snapshots, internal governance records, credentials, personal data and unrelated YiYuan assets are excluded.

9. Final Status

Field	Status
Technical report	PASS locally; five pages
Knowledge base	CONDITIONAL; metadata-only and incomplete
Agent execution	PASS locally
Scenario test	PASS locally; authorization not granted
Auditability	CONDITIONAL; local evidence only
Official submission route	EMAIL_ROUTE_VERIFIED; SUBMISSION PENDING
Final submission	EMAIL_PENDING

This report is a submission candidate with a reviewer-accessible public repository, not an official acceptance or submission receipt. The authorized email remains pending; the S1-S5 scenario authorization boundary remains explicitly disclosed.